2. DEFLECTION OF STATICALLY DETERMINATE STRUCTURE

General

The computation of deflection is of great importance in theory of structures due to:

(i) The analysis of indeterminate structures requires the development of displacement compatibility equations in addition to those of equilibrium equations.

(ii) Structures have to fulfill, in addition to strength criteria, certain specified requirements of rigidity to avoid excessive deformations under service loads.

There are several methods for computation of displacements of structures. In this section, two methods namely conjugate beam and the virtual work methods of deflection computation for statically determinate structures will be considered

Conjugate Beam Method

The conjugate beam method can be regarded as a special modification of the Moment – Area method for the computations of slope and deflection of beams. For the purpose of quick reference, the computation of slope and deflection of beams using the moment area method is based up on the following two moment area theorems, which were discussed in strength of materials and restated for easy reference:

Theorem 1

The difference in slope between any two points in a beam subjected to bending moment is equal to the area of the M/EI diagram between those points.

Theorem 2

The distance measured vertically between one point on a beam and the tangent from another point is equal to the first moment of the M/EI diagram between these two points, about the point where the displacement is measured.

As stated in the above theorems, the rotations and displacements at particular point of a beam cannot be obtained directly. Rather the results of the theorems are relative values and it requires making reference of suitable point and properly sketched elastic curve of the beam to obtain the actual displacement and rotation using the relative tangential deviations and relative difference in slopes. This is the basic limitation of the moment area method for the computation of slope and deflection of beams.

The conjugate beam method is therefore basically targeted to avoid this limitation of the moment are method and develop a procedure in which it will be possible to compute the actual slope and deflection at any required point of the beam. In this approach the concept of substitute or conjugate beam (a corresponding fictitious or analogous beam with certain support conditions) loaded with the angle change (M/EI) gives results for shear and moment of conjugate beam mathematically identical with the values of slopes and deflections of the elastic curve of the actual beam at any required point.

Thus the analogy or duality of parameters for the conjugate and real beams is indicated as follows:

Conjugate Beam		**Real Beam**
Shear	⟷	Slope
Moment	⟷	deflection

The practical statements for the conjugate beam method for the computation of slope and deflection of beams can be summarized in the form of the following two theorems:

Theorem1

The slope at any section of a loaded beam, relative to the original axis of the beam, is equal to the shear in the conjugate beam at the corresponding point where the conjugate beam is loaded by the M/EI diagram of the real beam.

Theorem 2

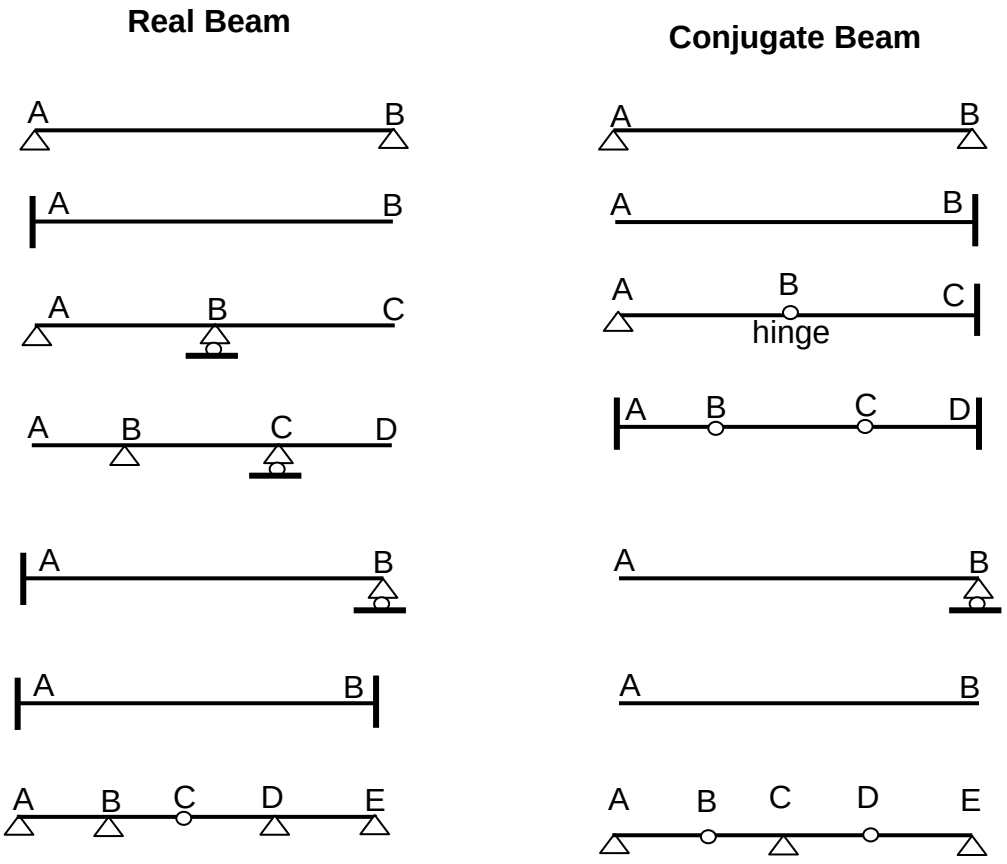
The deflection at any given section of a loaded beam, relative to the original position, is equal to the bending moment at the corresponding point of the conjugate beam where the conjugate beam is loaded by the M/EI diagram of the real beam.

The ordinate of the shear and bending moment diagram of the conjugate beam give, respectively, the slope and deflection at corresponding location in the real beam, measured with respect to the original un deflected position. The convention used for the sign of the load, shear and bending moment is the usual beam convention where the upward loads, indicating plus M/EI ordinates, are considered positive.

Although it is difficult to state simple rule to cover the relationships for all types between the real beam and the conjugate beam, the following observations may help for the proper substitute of conjugate beams for all ordinary cases.

a) The span of the conjugate beam is equal to the span of the real beam
b) A simple support in the real beam transforms in to simple support in the conjugate beam.
c) A fixed support in the real beam becomes a free end in the conjugate beam and vice versa
d) An interior support in the real beam becomes a hinge in the conjugate beam and vice versa.

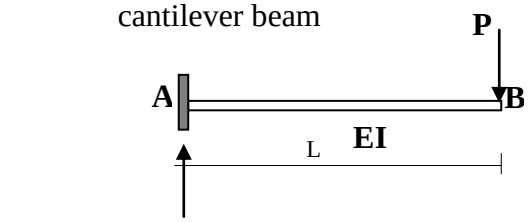
Based on the above facts, the following correspondence for some of the most common beam types and the associated conjugate beams are shown below.

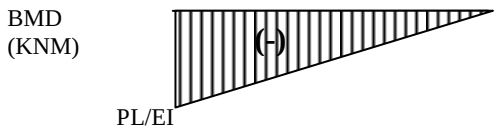


As indicated in the diagrams above, the conjugate beams may seem to be unstable or not adequately supported for some cases. However they are not to be subjected to any arbitrary load, rather the corresponding load effects of M/EI diagram of the real beam is found to maintain stable equilibrium condition of the seemingly unstable conjugate beam.

Examples

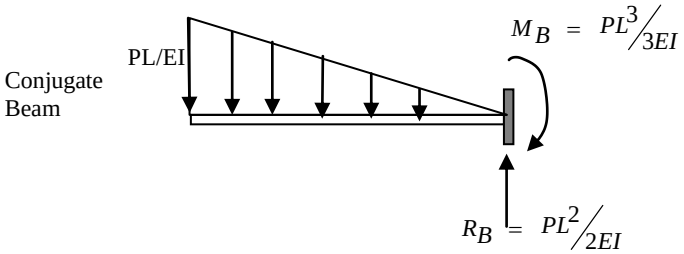
1. Determine the deflection and rotation at the free end for the following cantilever beam



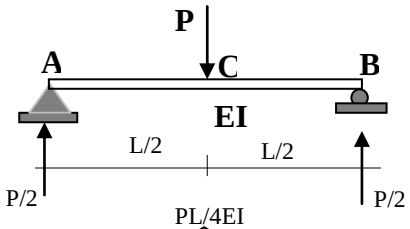


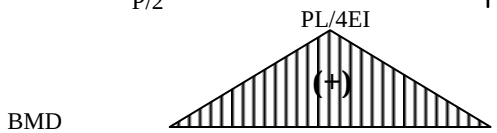
$$\theta_B = V_B = -PL^2/2EI \ , \ Clockwise$$

$$v_B = M_B = -PL^3/3EI \ , \ downward$$

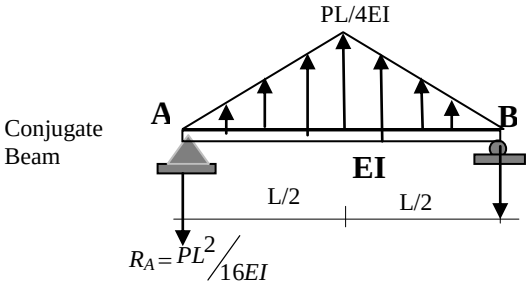


2. Determine the deflection at mid-point C and rotation at support A for the simply supported beam under the effect of concentrated load at the center.





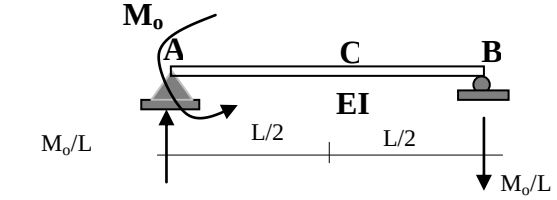
$$\theta_A = V_A = -PL^2/16EI \ , \ Clockwise$$

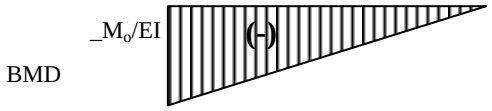


$$v_C = M_C = -PL^3/48EI \ , \ downward$$

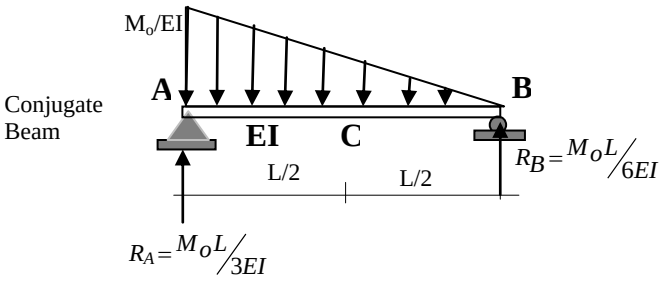
$$M_C = \frac{1}{2}\left(\frac{PL}{4EI} * \frac{L}{2}\right) * \frac{L}{6} - \frac{PL^3}{16EI} * \frac{L}{2} = \frac{PL^3}{48EI}$$

3. Determine the deflection at mid-point C and rotation at support A for the simply supported beam under the effect of concentrated Moment at support A





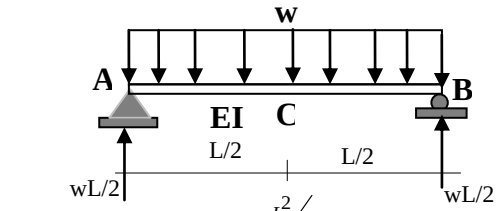
$$\theta_A = V_A = \frac{M_o L}{3EI} \text{ , Counter clock wise}$$

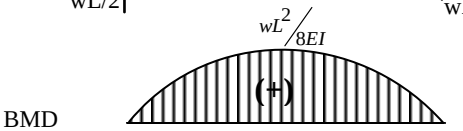


$$v_C = M_C = \frac{7M_o L^2}{48EI} \text{ , Upward}$$

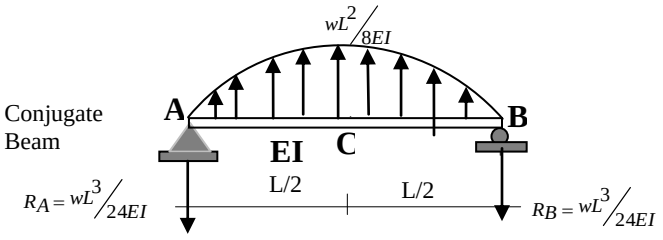
$$M_C = \frac{M_o l^3}{6EI} * \frac{L}{2} - \frac{1}{2}\left(\frac{M_o}{2EI} * \frac{L}{2}\right) * \frac{L}{6} - = \frac{7M_o L^2}{48EI}$$

4. Determine the deflection at mid-point C and rotation at support A for the simply supported beam under the effect of uniformly distributed load





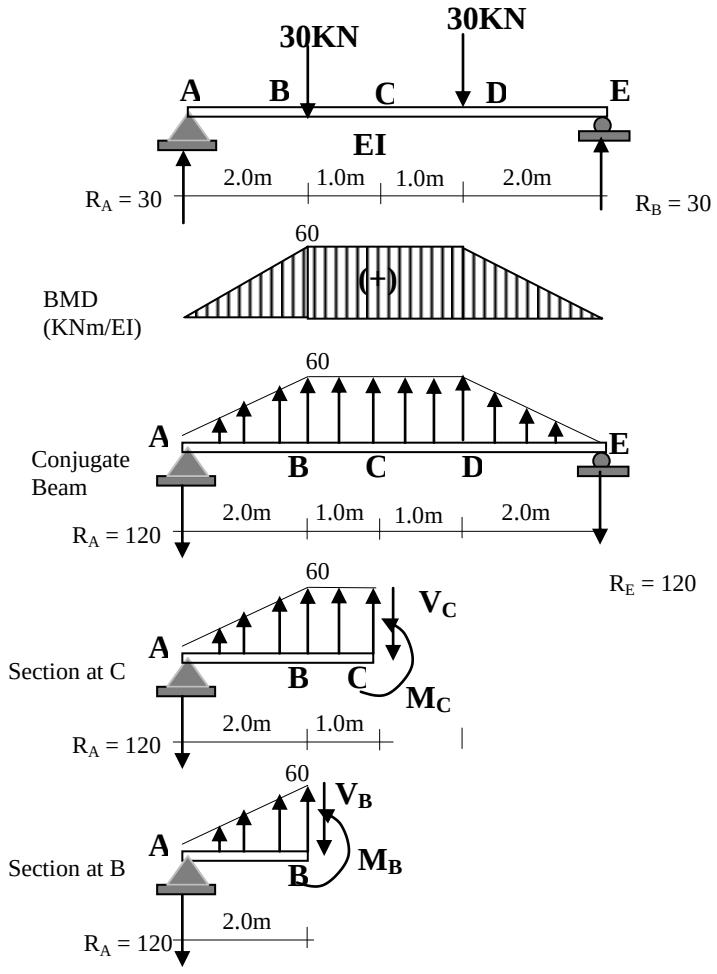
$$\theta_A = V_A = -\frac{wL^3}{24EI} \text{ , clock wise}$$



$$v_C = M_C = -\frac{5wL^4}{384EI} \text{ , downward}$$

$$M_C = \left(\frac{2}{3} * \frac{wL^2}{8EI} * \frac{L}{2}\right) * \frac{3}{8} L/2 - \frac{wL^3}{24EI} * \frac{L}{2} = -\frac{5wL4}{384EI}$$

5. Using conjugate beam method, determine the deflection at C and the slopes at A and B for the following beam



$$M_C = \frac{1}{2EI}(60*2)*\left(1+\frac{2}{3}*2\right) + \frac{60*1^2}{2\#I} - \frac{120*3}{EI} = \frac{-230}{EI}$$

$$V_B = \frac{60*2}{2EI} - 120 = \frac{-60}{EI}$$

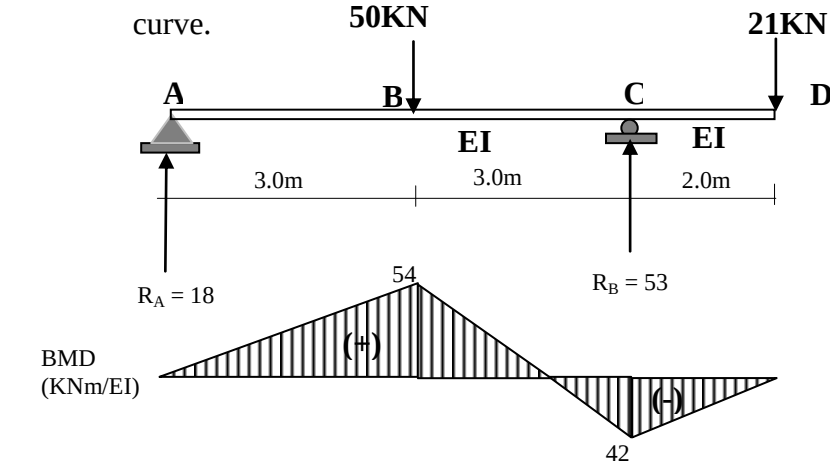
$$V_A = \frac{-120}{EI}$$

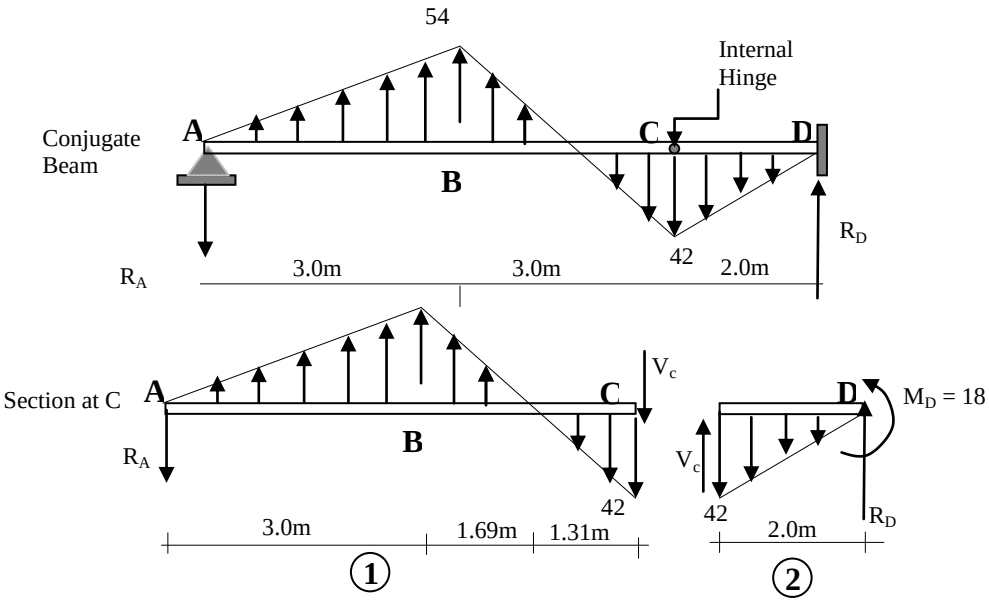
$$\theta_A = V_A = -120/EI \ , \ Clockwise$$

$$\theta_B = V_B = -60/EI \ , \ Clockwise$$

$$v_C = M_c = -230/EI \ , \ downward$$

6. Using conjugate beam method, determine the displacement at B and D and the slope at A and C for the following beam. Sketch the corresponding elastic curve.





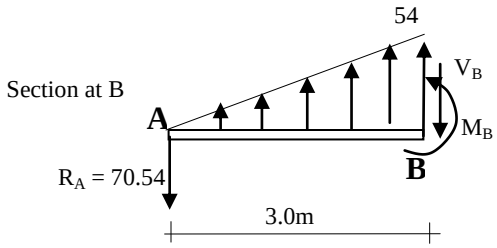
① $\circlearrowleft+ \sum M_A = \frac{54*3}{2EI}*\left(\frac{2}{3}*3\right) + \frac{54*1.69}{2EI}*\left(3+\frac{1}{3}*1.69\right) - \frac{42*1.31}{2EI}*\left(3+1.69+\frac{2}{3}*1.31\right) - V_C*6 = 0 \rightarrow V_C = 28.58$

$\sum F_y = 28.54 - R_A = 0 \rightarrow R_A = 70.54$

② $\circlearrowleft+ \sum M_D = \frac{42*2}{2EI}*\left(\frac{2}{3}*2\right)*\left(1+\frac{2}{3}*2\right) + -28..58*2 + M_D = 0 \rightarrow M_D = 1.174$

$\sum F_y = 28.58 - \frac{42*2}{2} + R_D = 0 \rightarrow R_D = -13.42$

Taking Section at B and considering the left part



$\circlearrowleft+ \sum M_B = \frac{70.54*3}{EI} - \left(\frac{54*3}{2EI}\right)*\frac{3}{3} + M_B = 0 \rightarrow MB = -\frac{130.62}{EI}$

$v_B = M_B \;;\; v_D = M_D \;;\; \theta_A = V_A \;;\; \theta_C = V_C$

$v_B = -\frac{130.62}{EI}$, *downward* $\qquad \theta_A = -\frac{70.54}{EI}$

$v_D = +\frac{1.174}{EI}$, *upward* $\qquad \theta_D = +\frac{28.58}{EI}$

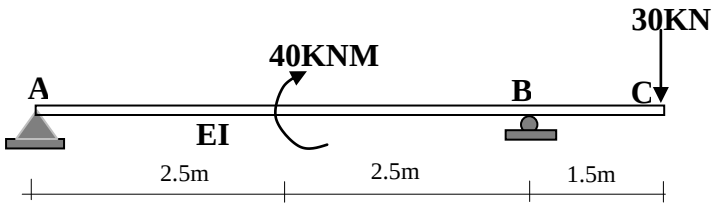
Note that EI is to be measured in KNM^2.

2

Exercise

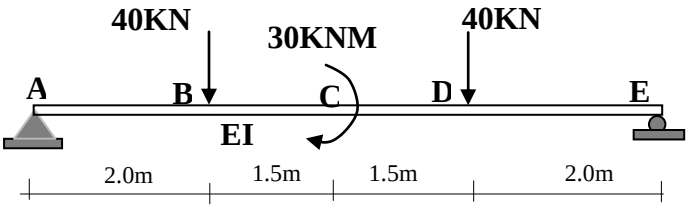
Using conjugate beam method, determine the rotation and deflection at the specified points for the following beams.

(a)



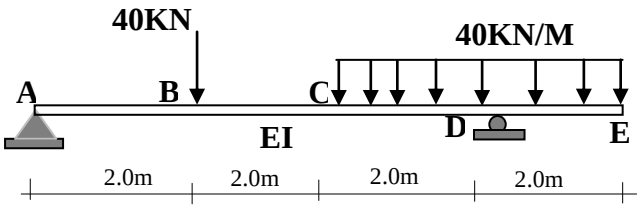
v_C = ?
θ_A = ?
θ_B = ?

(b)



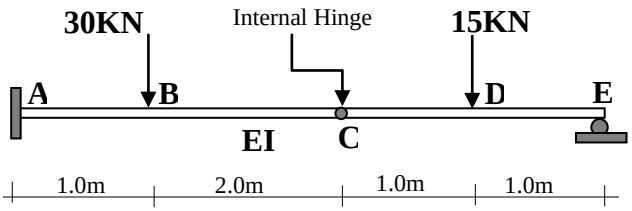
v_C = ?
θ_B = ?
θ_E = ?

(c)



v_E = ?
θ_B = ?
θ_D = ?

(d)



v_c = ?
θ_E = ?